

PySpark Wine Reviews: End-to-End Big Data ML

A Scalable Big Data Lifecycle Demonstration

DS410 - Society of Wine Specialists

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SWS

Project Overview &
Problem Definition

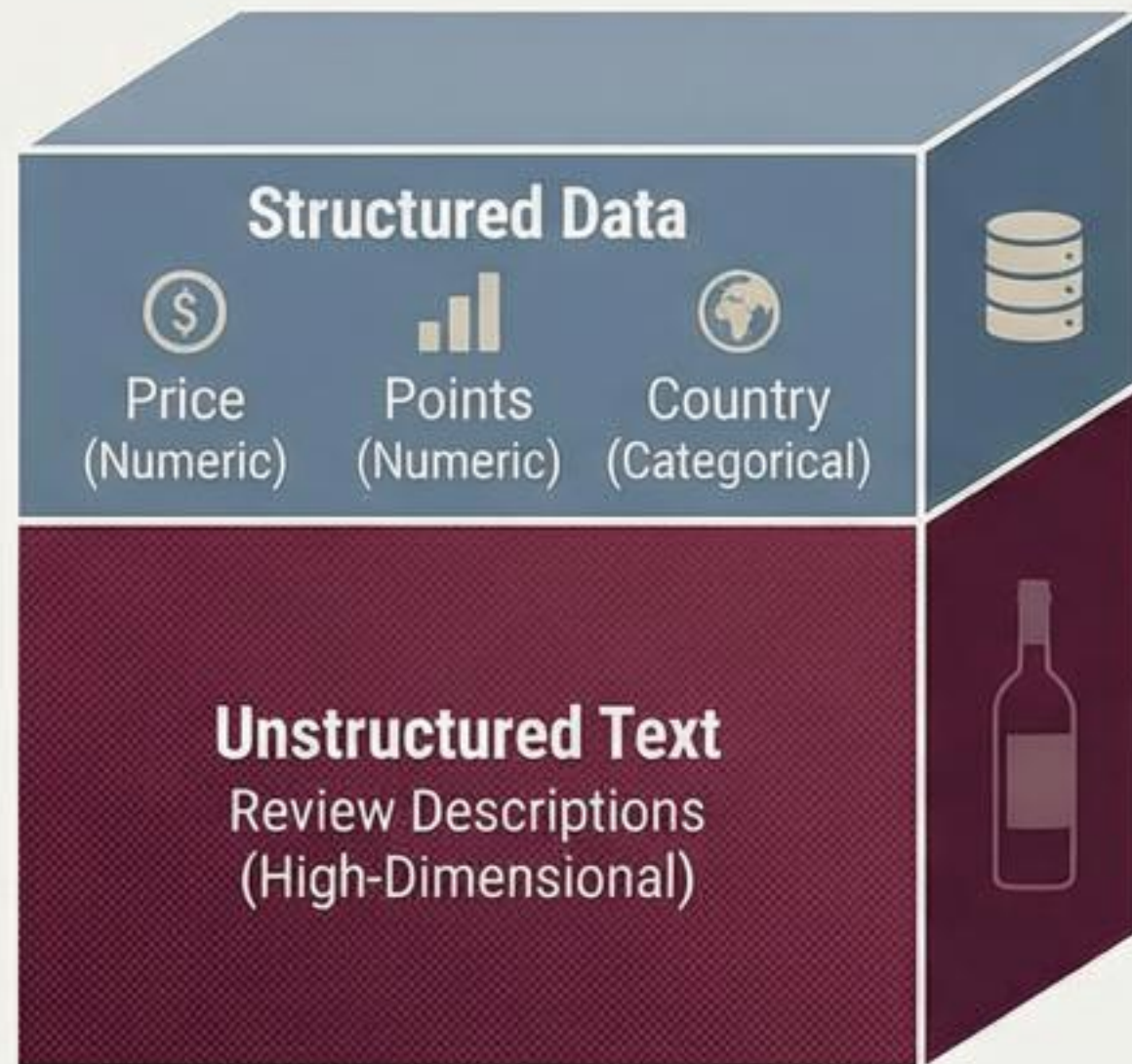
How accurately can we predict high end consumer goods?

The objective would be to fine tune a model to be able to make predictions for high end consumer goods in general, in this project we used a wine dataset to attempt to accurately predict consumer taste using the data sets rating by point system.

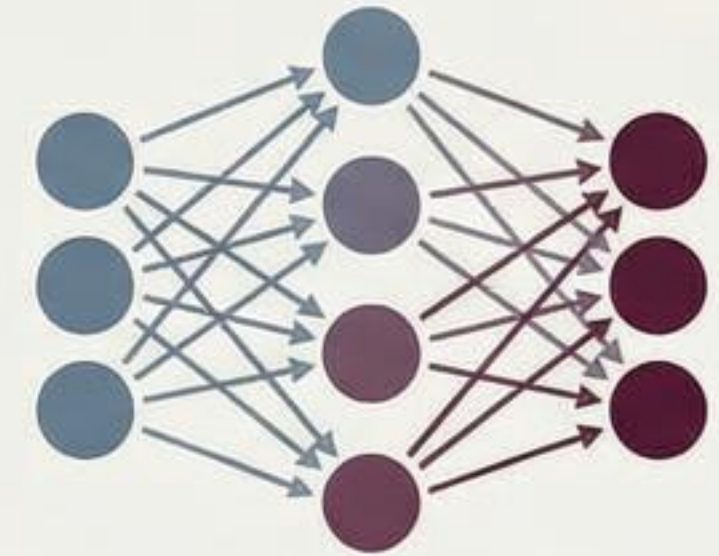
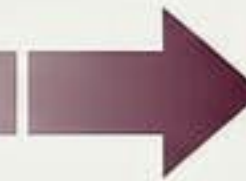
Dataset Characteristics & The Big Data Challenge

Source: WineEnthusiast Reviews (Kaggle) | Dimensions: ~130,000 Observations

Data Source & Composition



The Big Data Bottleneck & Solution



Traditional Libraries (e.g., Pandas)

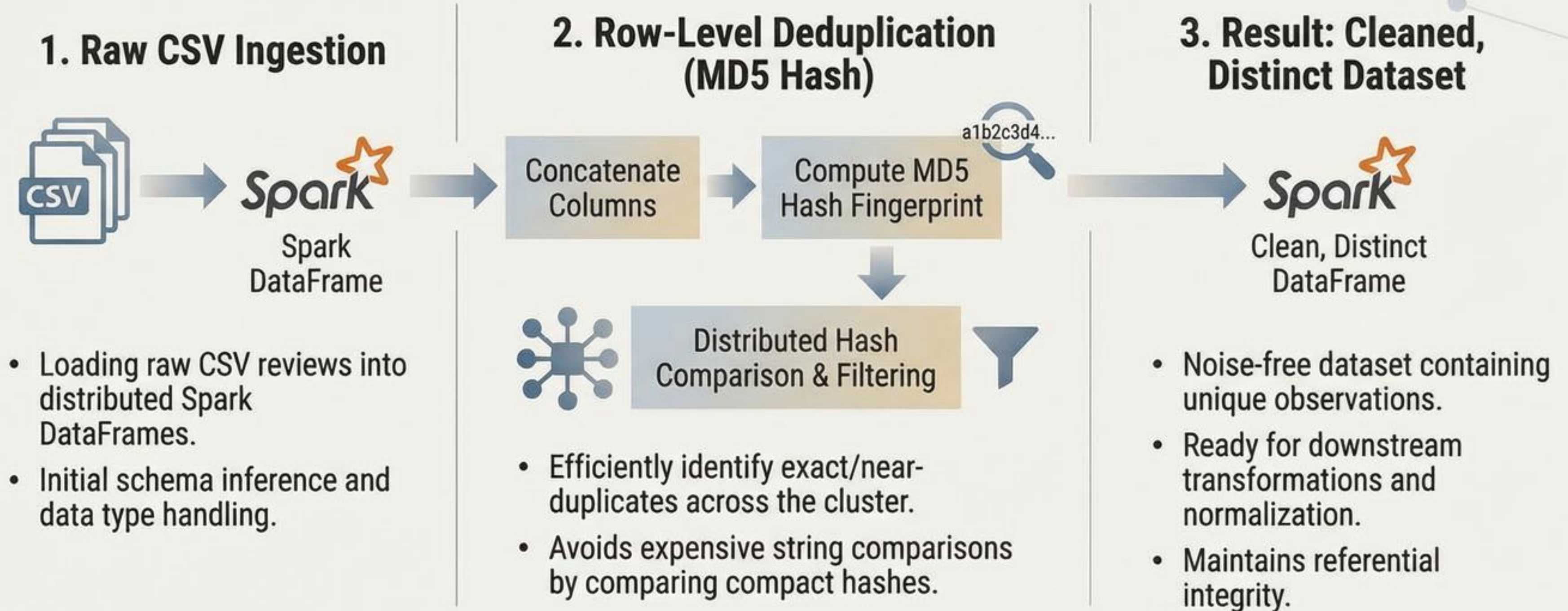
Struggles with memory overhead for 130k+ text documents during tokenization

Distributed Approach (PySpark)

Leverages Spark RDDs & ML Pipelines for parallel storage and computation across cluster nodes, preserving semantic richness.

Ingestion & Deduplication Strategy: Ensuring Data Integrity

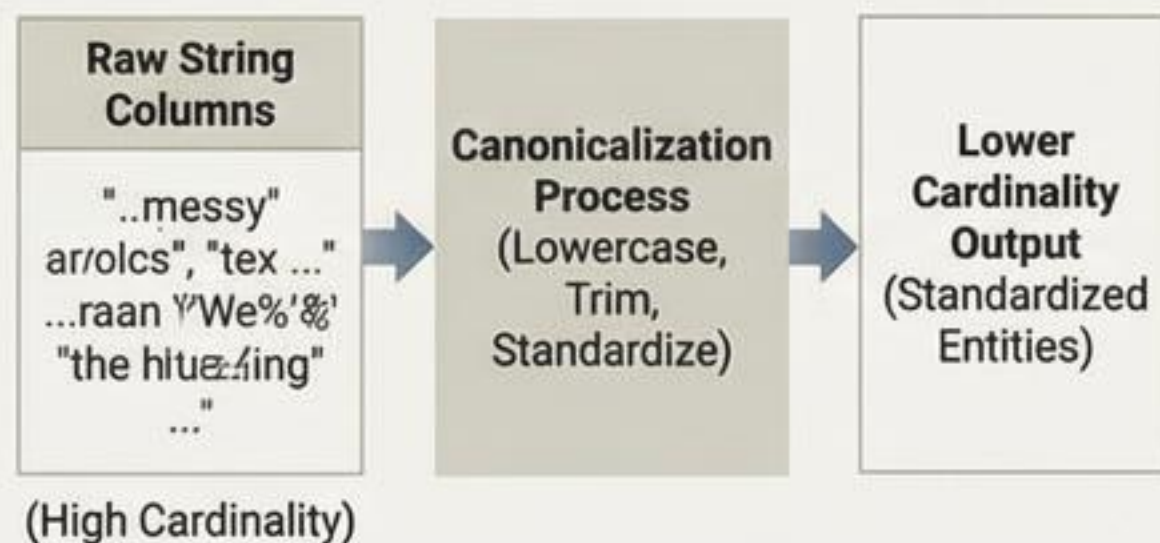
Speaker: Oluwatomisin Olaore | Technical Tone: Data Quality & Cleaning Algorithms



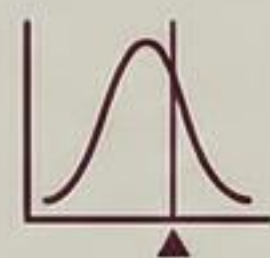
Data Cleaning & Normalization Strategy



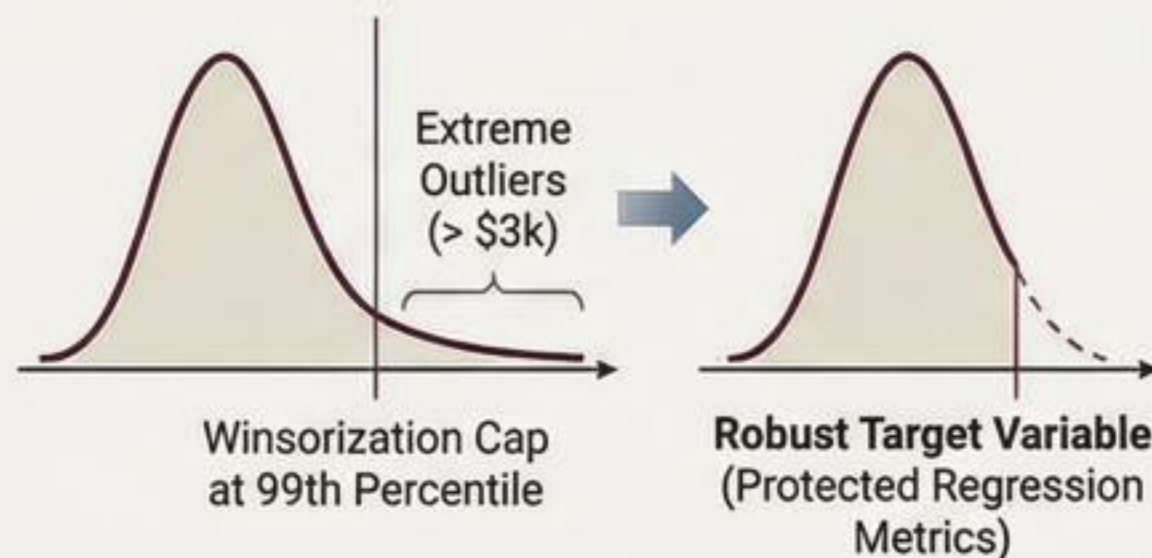
String Canonicalization



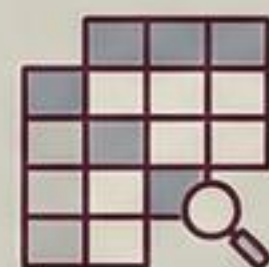
String columns are canonicalized to lower cardinality.



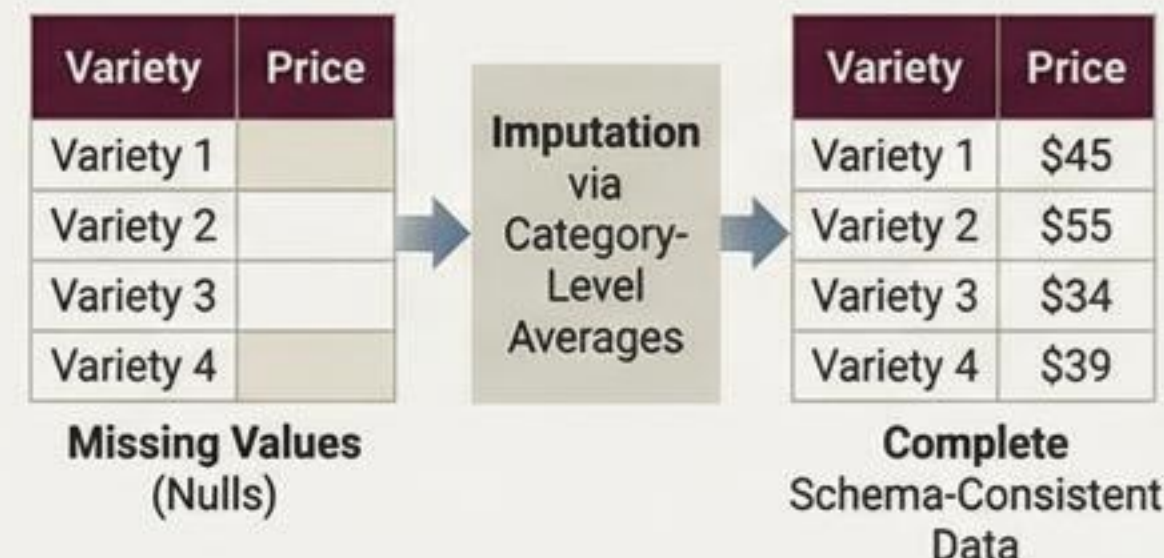
Numerical Noise Handling (Winsorization)



Extreme price outliers above the 99th percentile are Winsorized.



Missing Value Imputation

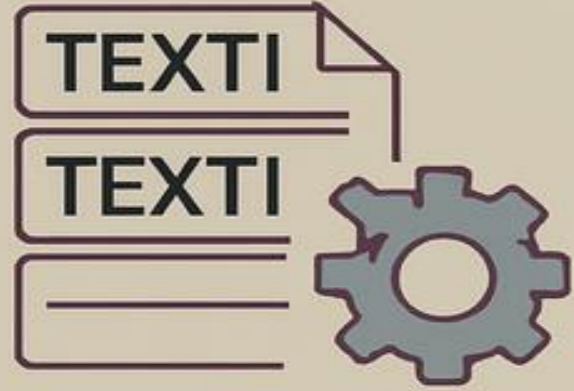


Missing values are imputed with category-level averages.

Outcome: A noise-reduced, schema-consistent DataFrame optimized for scalable modelling.



Data Cleaning Steps



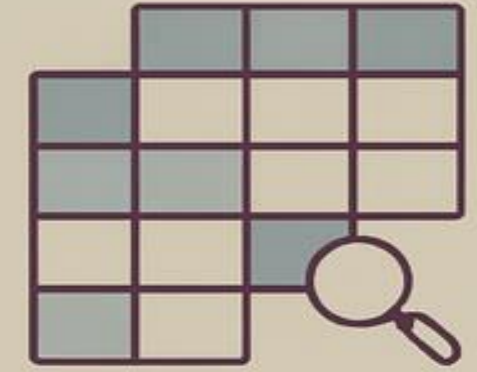
Automatically search for the CSV, set *multiLine=True*

Drop noise columns

Remove columns like "Unnamed: 0" or "index"

Correct data types

- points → int
- price → double



Produce null report

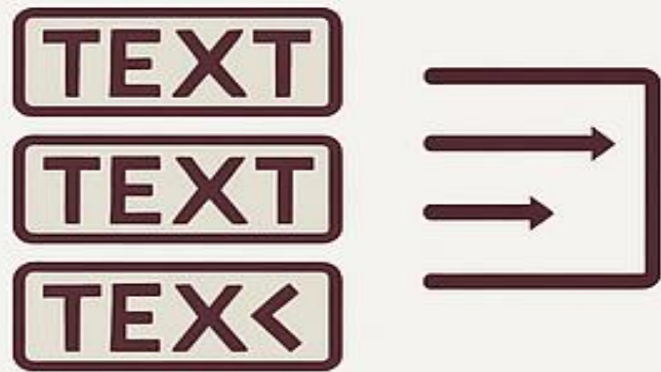
Outcome: A noise–reduced, schema–consistent DataFrame optimized for scalable modelling



Data Cleaning Approach

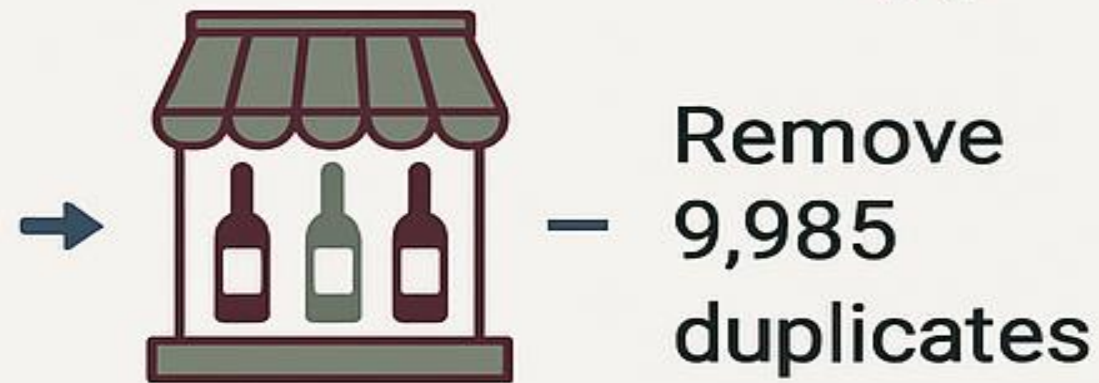
Text Normalization

- Remove accents
- Convert to lowercase
- Collapse whitespace



Fingerprint Deduplication

- Normalize description
- Compute SHA-256 hash



Missing Value Handling

- Drop missing description
- Drop missing variety



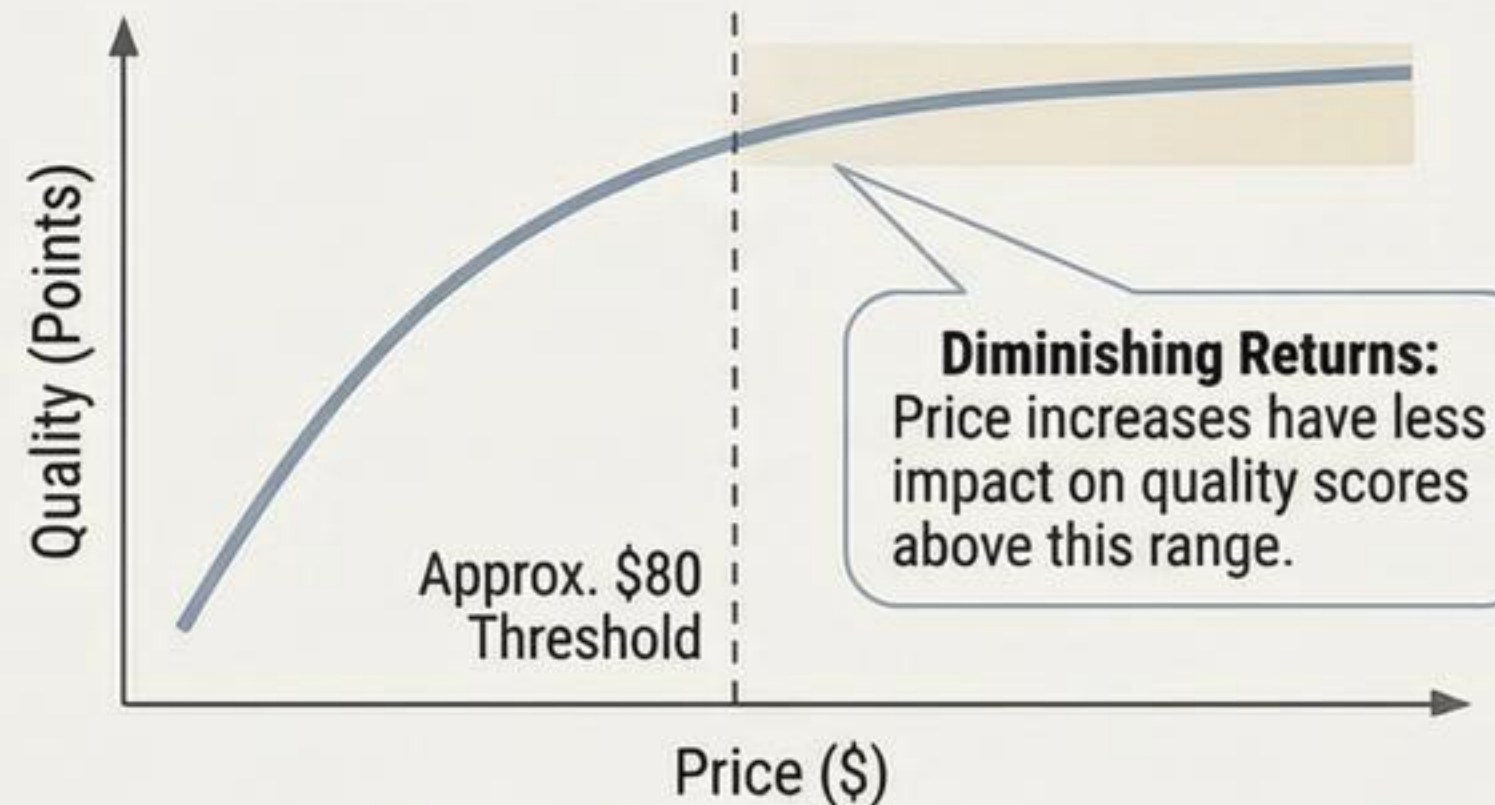
Outcome: Improved data consistency suitable for downstream modelling



Feature Correlation Insights: Price, Quality & Textual Complexity

Diminishing Returns & The Necessity of NLP

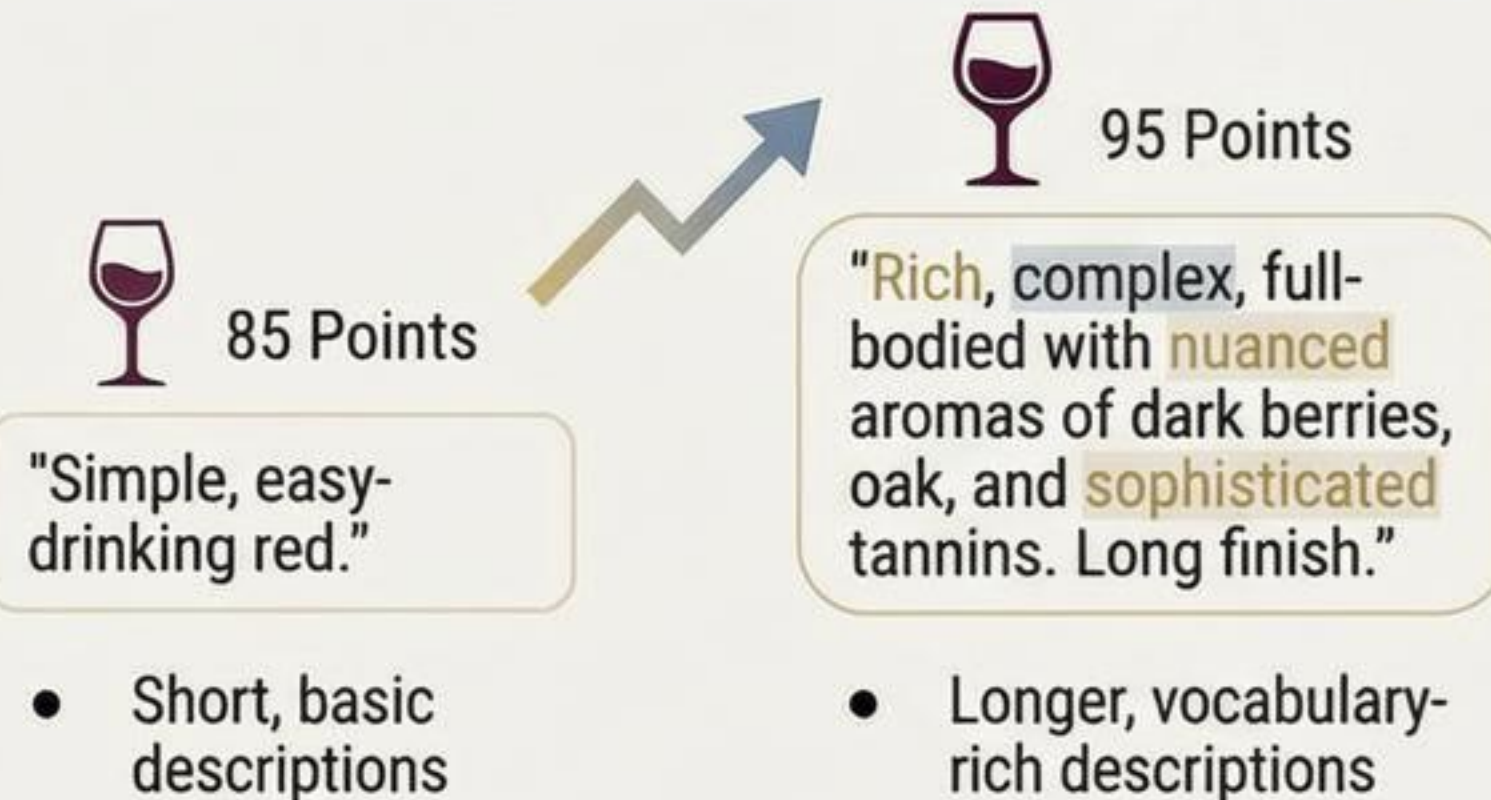
Price vs. Quality Correlation



Observed diminishing returns at high price points, indicating price is not a linear predictor of quality.



Textual Complexity & Rating



Higher-rated wines tend to have longer, more complex descriptions, justifying investment in NLP.

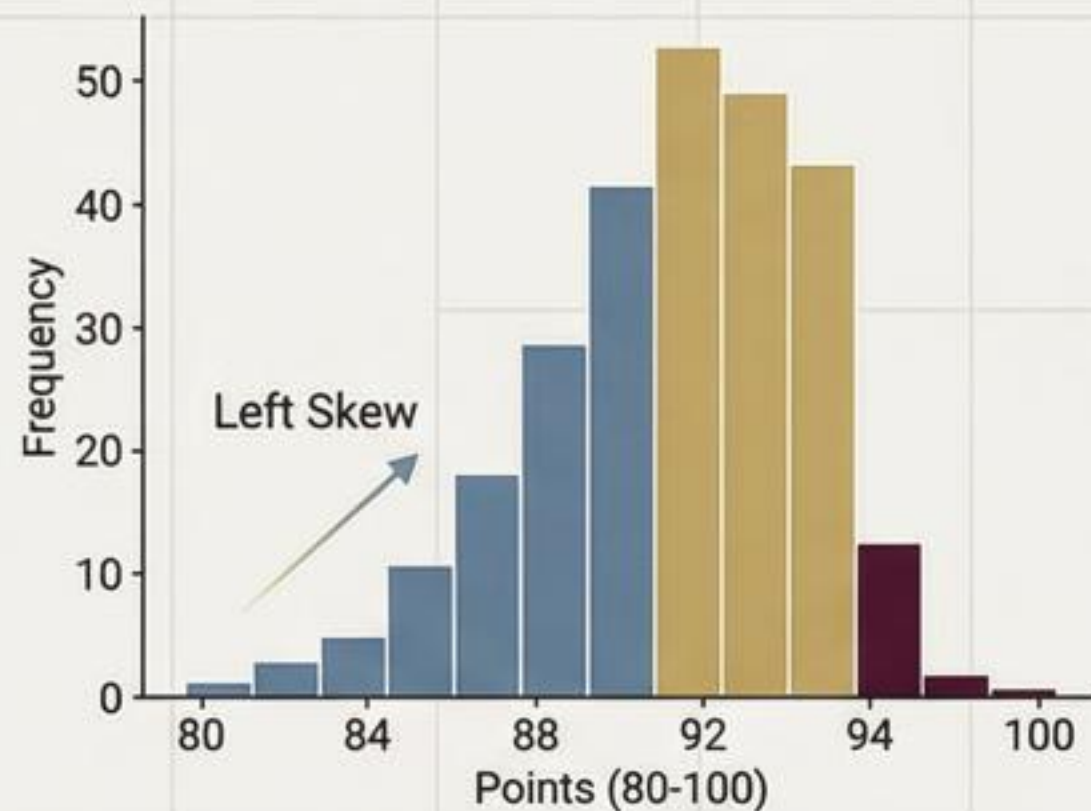


Conclusion: Validates the need for sophisticated NLP features (TF-IDF, PCA) to capture semantic nuance that numeric fields alone cannot express.



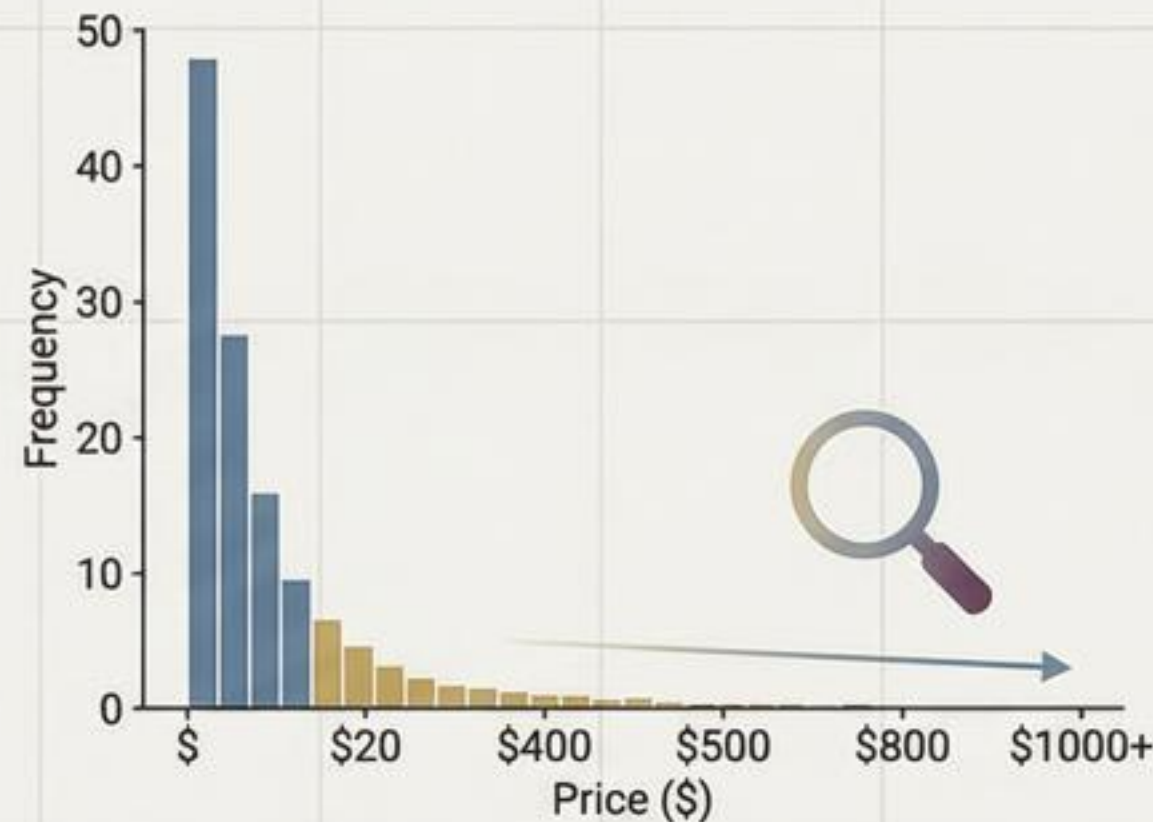
Exploratory Data Analysis: Statistical Distributions & Imbalance

Points Distribution (Target Variable)



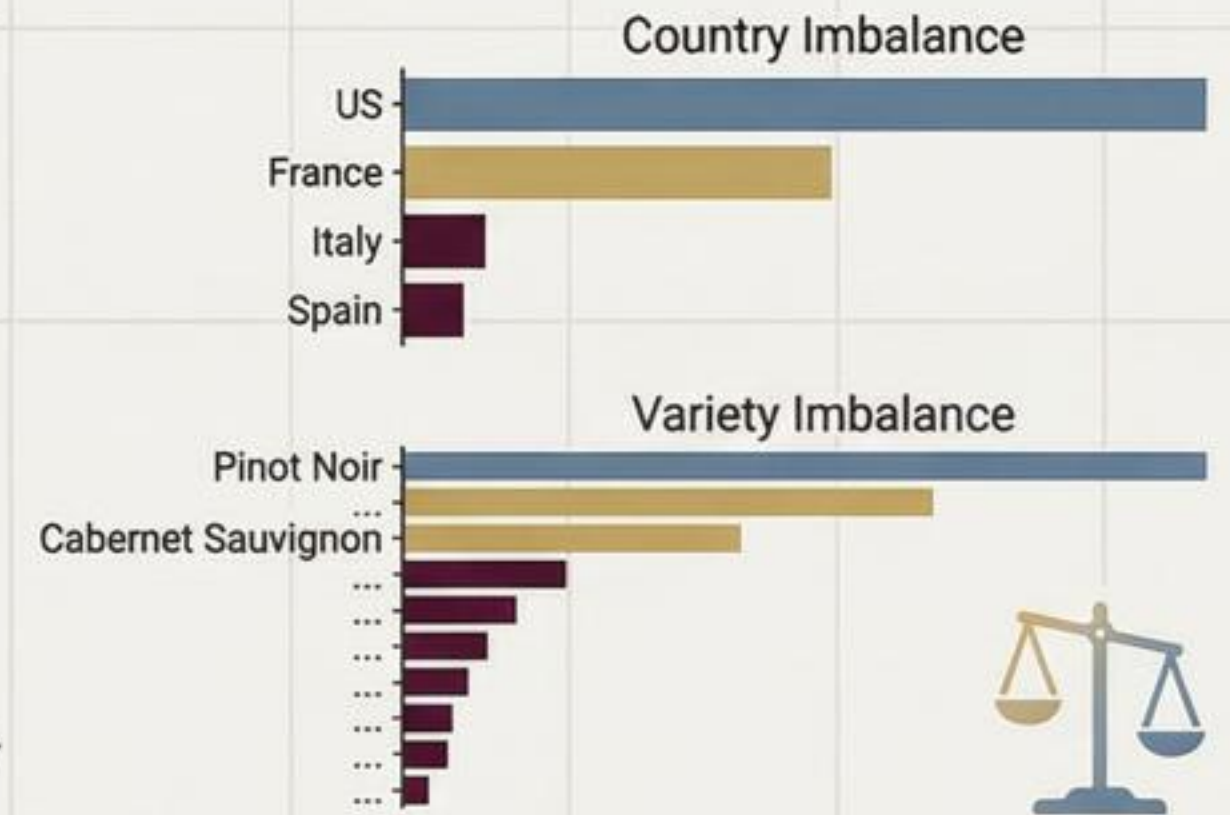
Concentrates between 80-100 with mild left skew. Most wines rated 88-94 points.

Price Distribution (Target Variable)



Follows a long-tail distribution. High frequency of low prices, with few extreme outliers.

Categorical Imbalance (Country & Variety)



Severe class imbalance, dominated by US and France. Guiding the choice of F1-score over accuracy for multiclass evaluation.

Data Analysis — Why These Five Questions?

1. Country Distribution

To understand the global footprint of wine production and identify dominant regions that shape most of the dataset.

2. Price & Score Statistics by Country

To evaluate how different regions position themselves in the market and whether price/quality patterns vary meaningfully across countries.

3. Grape Variety Distribution

To capture product-level diversity, reveal head-heavy vs. long-tail categories, and understand how varietal structure

2. Price & Score Statistics by Country

To evaluate how different regions position themselves in the market and whether price/quality patterns vary meaningfully across countries.

4. Vintage Trends

To examine how wine characteristics and review coverage evolve over time, and to separate real temporal patterns from dataset-driven artifacts.

5. Score-Price Relationship

To quantify how critic evaluations transiate into market value and assess whether higher scores consistently predict higher prices.

Why Local Mode Struggled & How Roar-Collab Helps

Why was this problem intractable for local mode?

The pipeline generates high-dimensional, wide intermediate features (TF-IDF + OneHot + assembled vectors).

Local mode is limited by single-machine memory/CPU, leading to unstable runs and frequent OOM during feature assembly and text processing.

End-to-end runtime grows quickly because ETL + text featurization + model training compete for the same limited resources..

How would ICDS Roar-Collab assist?

Roar-Collab provides distributed memory and parallel compute, allowing expensive Spark stages (TF-IDF, PCA, vector assembly) to scale beyond a laptop's limits.

The environment supports multi-node experiments, enabling us to empirically validate horizontal scalability (time-to-train₁, speed-up₁).

Large-scale storage workflows benefit from Parquet columnar format, reducing I/O overhead for repeated reads and filtered workloads.

What We Did on Roar-Collab

Setup

- Staged data on Roar and standardized Spark paths.
- Switched key outputs to Parquet.
- Ran a Slurm-managed multi-node Spark job.
- Used repartition + MEMORY_AND_DISK + unpersist for stability

Challenges

Core bottleneck; memory pressure from wide text/categorical feature vectors in cleaning/assembly.

Secondary: shuffle skew & I/O overhead
→ controlled repartition; Parquet + compression.

->Timeout

Methods

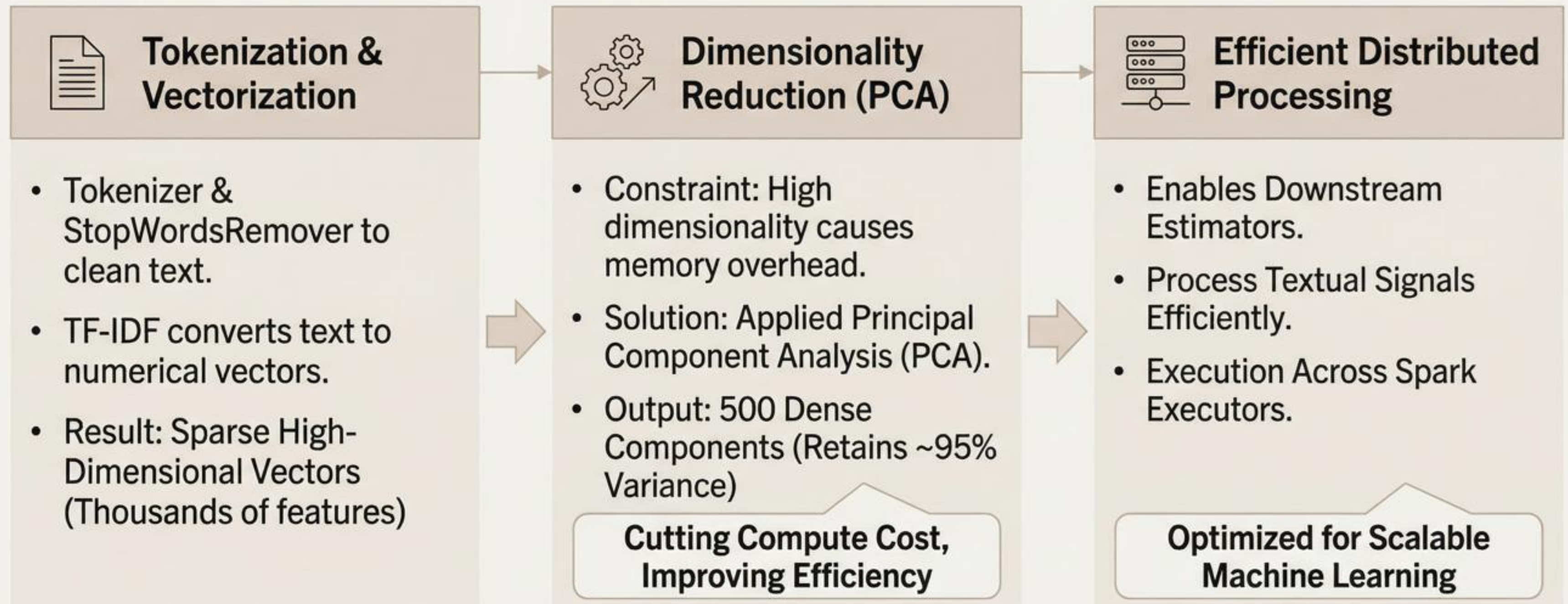
- Reproducible pipeline: cleaning → dedup → structured + text features.
- TF-IDF for interpretability, PCA for tractable scale.
- Unified ML-ready features for multi-task modeling.

Roar-Collab Usage Summary

- Conducted 6 total experiments for pipeline testing, feature assembly, and scaling behavior.
- Utilized both 2-node and 4-node Spark cluster configurations to compare how partitioning and PCA workloads behave under different resource allocations.
- Experiments covered ingestion → cleaning → TF-IDF/PCA → ML-ready feature generation.
- Multi-node jobs typically ran for several minutes to tens of minutes, depending on PCA dimensionality and the number of partitions.

Section 4: Feature Engineering & Model Architecture

Speaker: Ce Liang (Model Lead)



Feature Engineering & Model Architecture

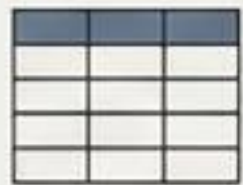
Speaker: Ce Liang | **Tone:** Highly Technical, MLOps

The Master Feature Vector

PCA Text Embeddings



One-Hot Encoded
Categorical Features



Concatenation



Master Feature Vector
(Spark ML Lib)



Master Feature Vector
(Spark ML Lib)

Serialize & Save
Fitted Transformers

Artifact Management & MLOps

Disk Storage
(outputs/models/)



Loaded Pipeline
(for Inference)

New Data

Loaded Pipeline
(for Inference)

Predictions/
Scores

✓ **Scoring without Retraining:**
Reproducible, Production-Grade
Feature Transformation.

PCA text embeddings are concatenated with one-hot encoded Country and Variety into a single feature vector.



Core Machine Learning Tasks (SWS: Scalability & Performance)

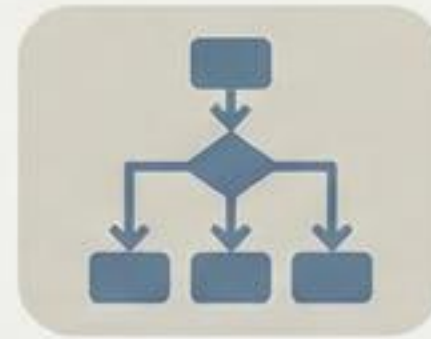
Showcasing PySpark MLlib's Versatility on Engineered Feature Sets

Task 1: Regression (Numeric Target)



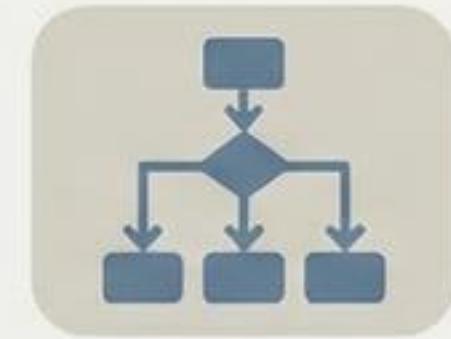
Models: Linear Regression, Random Forest.
Target: price_winsor (Winsorized Price).
Focus: Predicting continuous wine price.

Task 2: Classification A (Categorical Target)



Model: Multinomial Logistic Regression.
Target: variety_canonical (Wine Variety).
Focus: Multiclass classification of distinct wine types.

Task 3: Classification B (Categorical Target)



Target: country_norm (Origin Country).
Focus: Predicting wine origin country.



Engineered Feature Vector (from Master Pipeline)



All tasks leverage the same robust, processed features for efficient parallel training.

Price Prediction: Models & Results

1. Baseline Linear Model (Text + Score Only)

Shows that tasting notes + critic score contain real pricing signal ($R^2 \approx 0.275$).

Captures broad tiers (cheap vs. premium) but misses many structural drivers like region or variety.

2. Adding Metadata (Region, Variety, Vintage)

Location and grape information improve accuracy ($R^2 \rightarrow \approx 0.321$).

Still limited because linear models cannot capture non-linear interactions (e.g., Burgundy Pinot vs. generic blends).

3. Random Forest with Full Feature Set

Non-linear model handles complex interactions and boosts performance significantly.

Good at assigning wines into reasonable price tiers, even if exact prices remain hard due to unobserved factors.



Variety Classification (Multiclass)

1. Initial Attempt (Leakage Issue)

Reusing full feature vector accidentally included one-hot variety encoding → logistic regression reach ≈ 0.997 accuracy.

Performance was meaningless because model “saw” the label inside its own features.

2. No-Leakage Feature Redesign

Built a clean feature set: tfidf_pca text embedding, points, vintage_year and region one-hots only. Removes all variety-related columns so model must infer grape type from tasting notes + region patterns.

3. Realistic Model Performance

Multinomial logistic regression scores ≈ 0.572 accuracy and ≈ 0.508 F1.

High accuracy on major varieties, weaker on long-tail grapes — consistent with the difficulty of predicting among hundreds of classes.



```
23/11/27 23:17:22 WMLM-BRO-Scheduler: Broadcasting large task binary with size 9.7 Mib
[Logistic Regression (no leakage)] variety_canonical - accuracy = 0.572, F1 = 0.508
Sample predictions (true vs predicted variety IDs):

[Stage 1398:>                                     (0 + 1) / 1]
+-----+-----+-----+-----+-----+-----+
|country_norm|province_norm|variety_canonical|label_variety|pred_variety|points|vintage_year|
+-----+-----+-----+-----+-----+-----+
|NULL|NULL|ojaleshi|622.0|0.0|88|2012|
|NULL|NULL|tannat|60.0|3.0|86|2015|
```

Country of Origin Prediction

1. Initial Model (Proxy Leakage)

Using province + variety one-hots led to ≈ 0.998 accuracy because many provinces uniquely determine the country.

The model was not “reading” tasting language—it was simply recovering country from geographic proxies.



2. No-Leakage Redesign (Text-Only Features)

We removed all geographic fields to avoid leakage and built features_country_text using only tfidf_pca, points, and vintage_year.

Forces the classifier to infer country purely from tasting vocabular and basic wine attributes.



3. Realistic Performance & Interpretation

Logistic regression reaches ≈ 0.746 accuracy and ≈ 0.695 F1—much lower but truly meaningful.

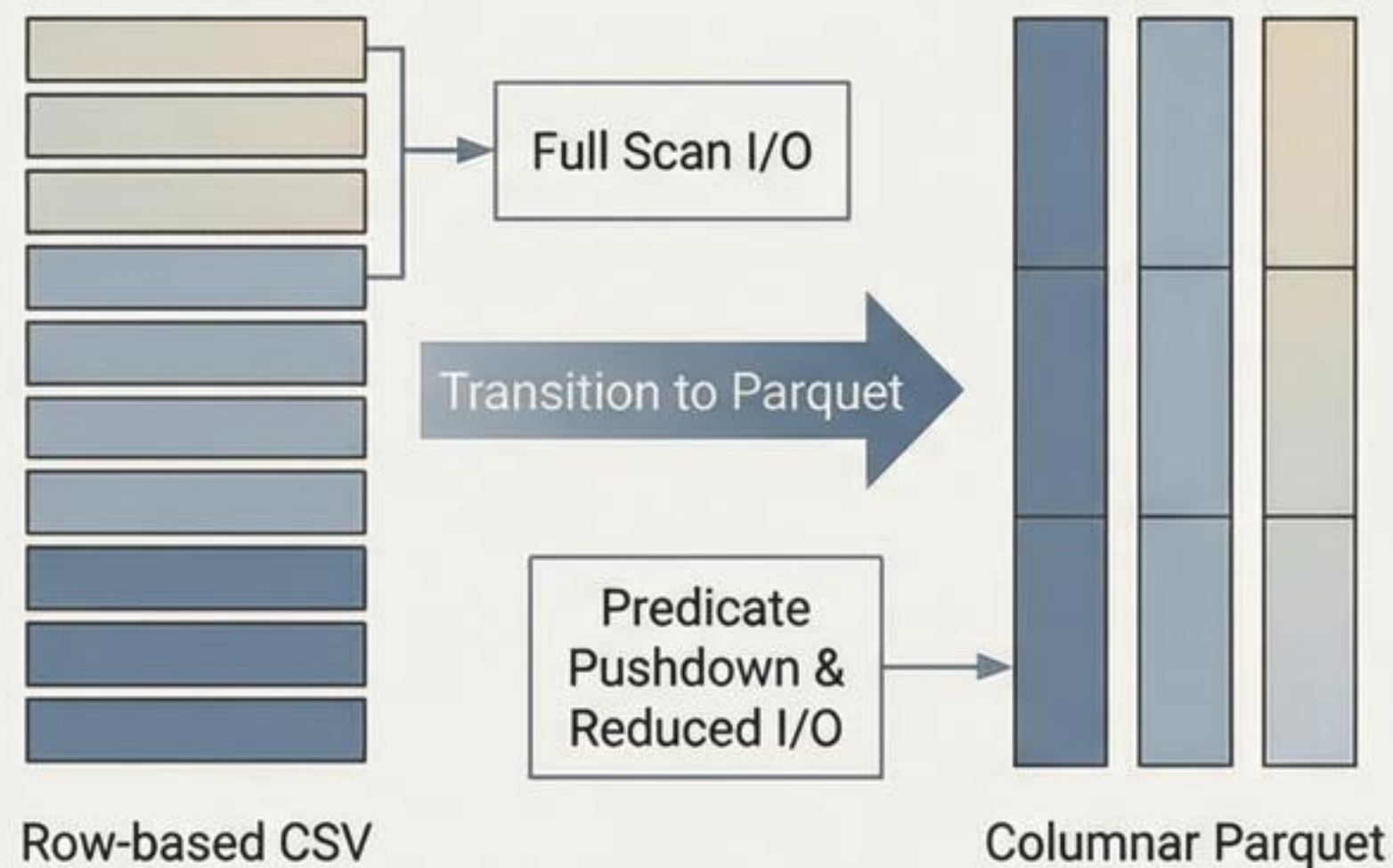
Text carries strong signals (distinct vocabularies across US / France / Italy / Spain / Argentina / Chile), yet confusion persists between stylistically similar countrys

[Logistic Regression (text-only, no leakage)] country_norm - accuracy = 0.746, F1 = 0.695

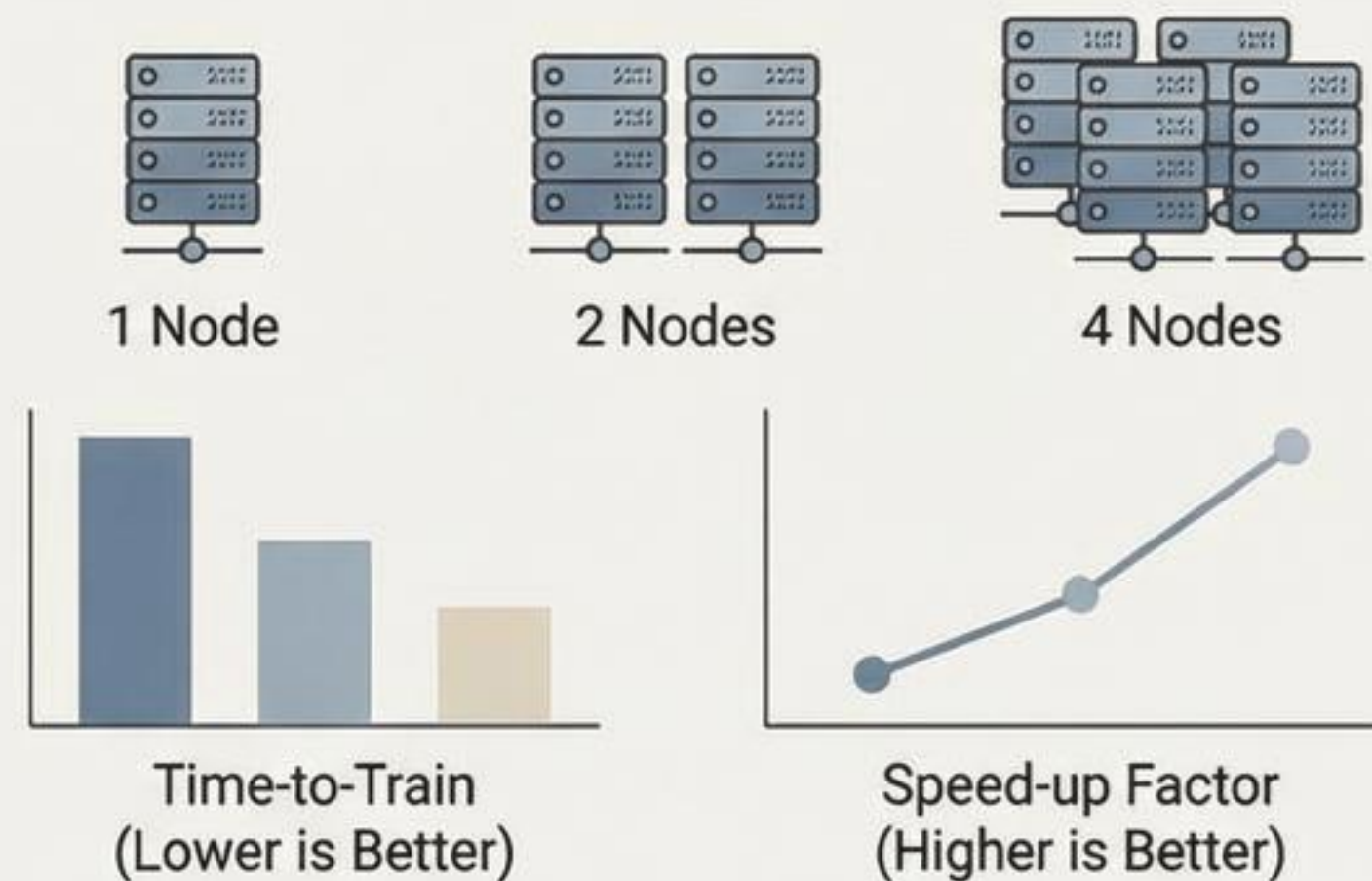
Sample predictions (true vs predicted country):

Infrastructure & Scalability Experiment

Storage Optimization: Parquet Columnar Format



Horizontal Scalability Test (410 Benchmark)



Goal: Empirically verify horizontal scalability by measuring reduced runtime with added hardware resources.

Evaluation Strategy: A Balanced Scorecard.

Assessing Predictive Power and Computational Cost

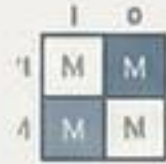
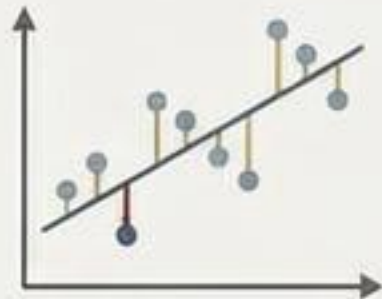


Regression Metrics

$\sqrt{x}$ RMSE (Root Mean Square Error)

$|A|$ MAE (Mean Absolute Error)

Gauges performance, with RMSE penalizing large errors more heavily.



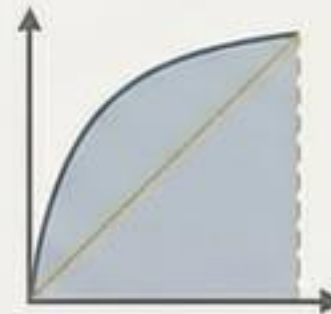
Classification Metrics



✓ Accuracy

 AUC (Area Under Curve)

Crucial for handling class imbalance; AUC provides a robust measure.

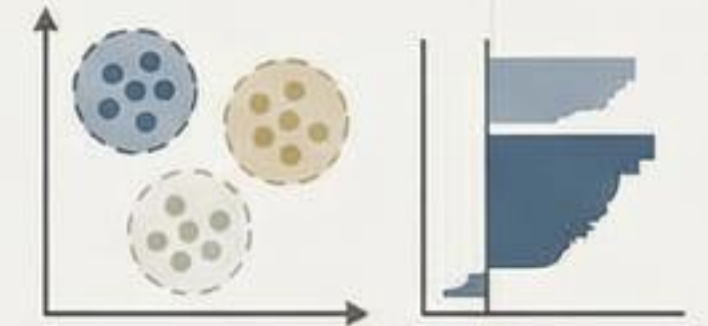


Clustering Metrics



Silhouette Score

Validates separation of taste profiles and cluster cohesion.



Predictive Power

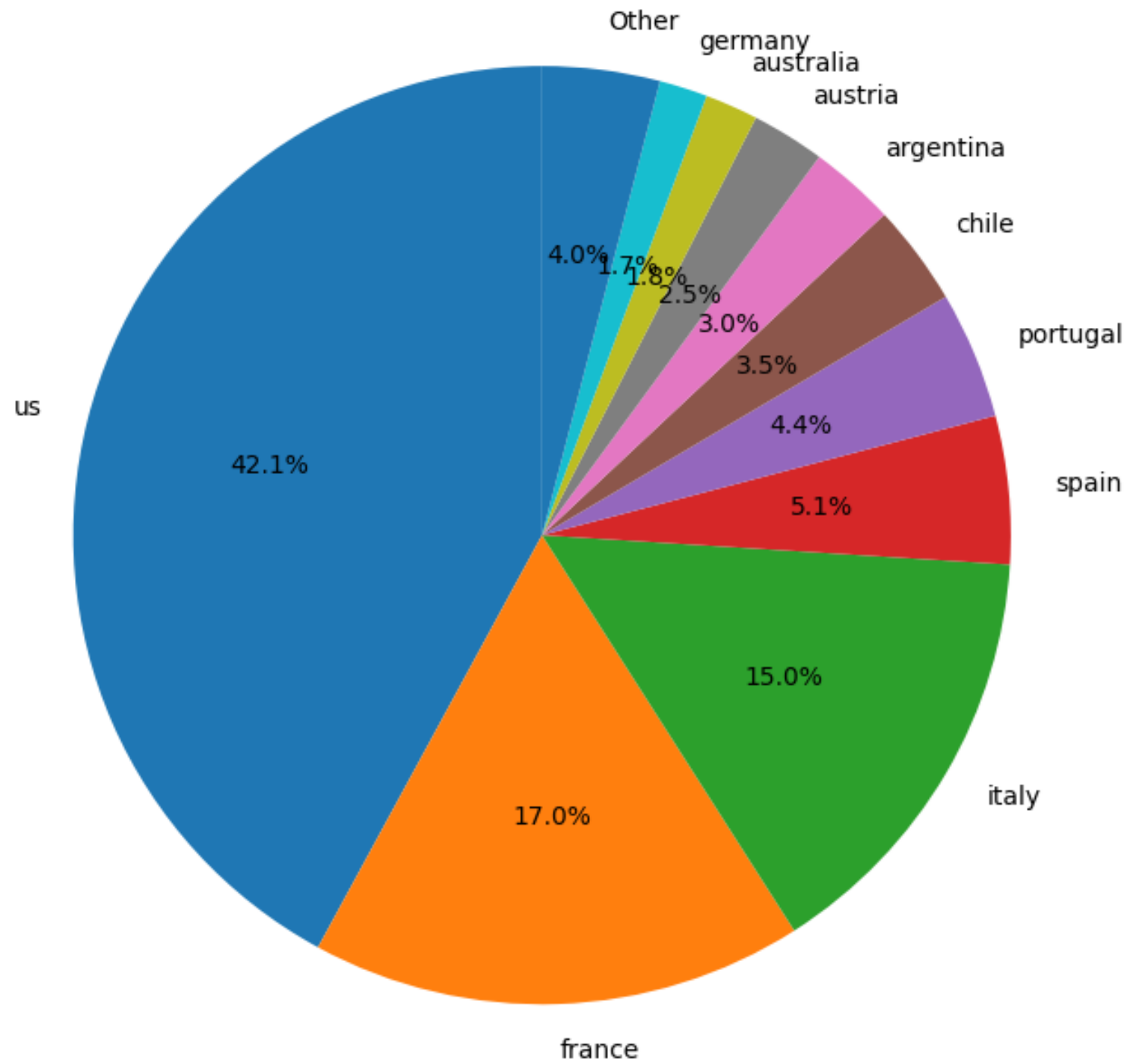


Computational Cost

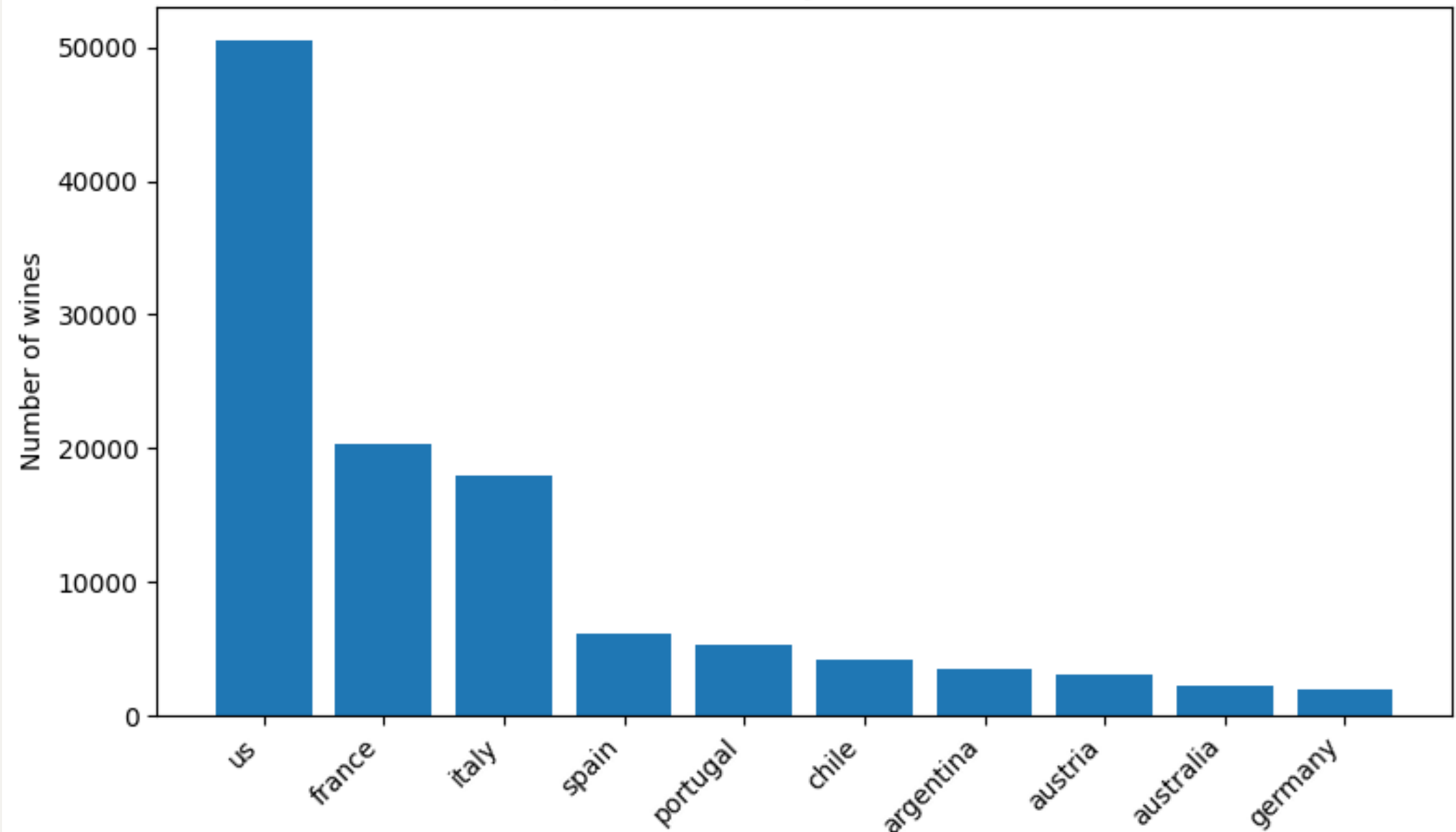


Comprehensive Project Assessment: Balancing model performance with resource efficiency

Share of wines by country (Top 10 + Other)

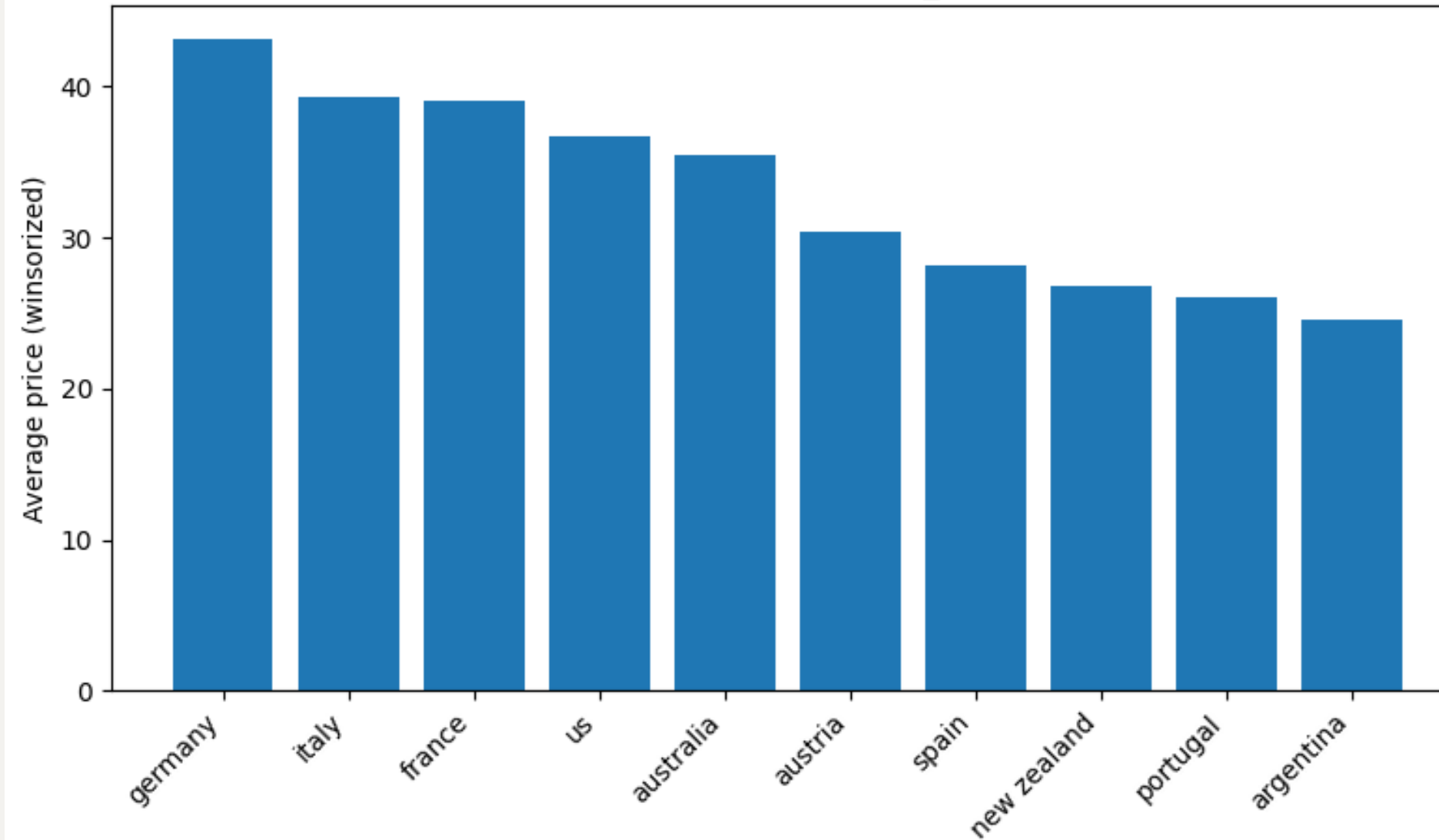


Top 10 countries by number of wines

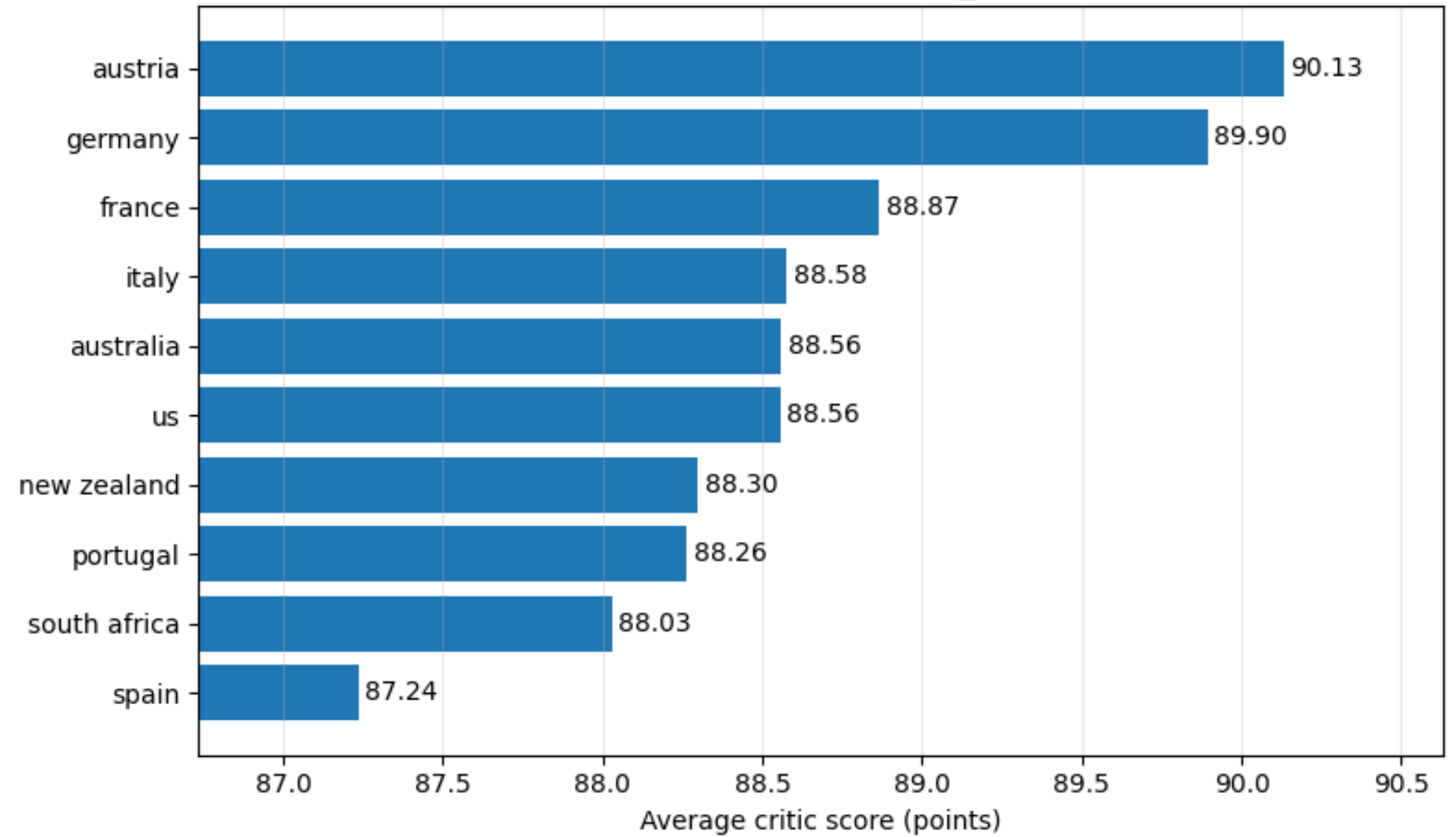


Distribution implies that any downstream models or analyses will primarily reflect patterns in US, French, and Italian wines, while conclusions about smaller producers should be interpreted with caution due to more limited sample sizes.

Top countries by average price (n_wines >= 500)

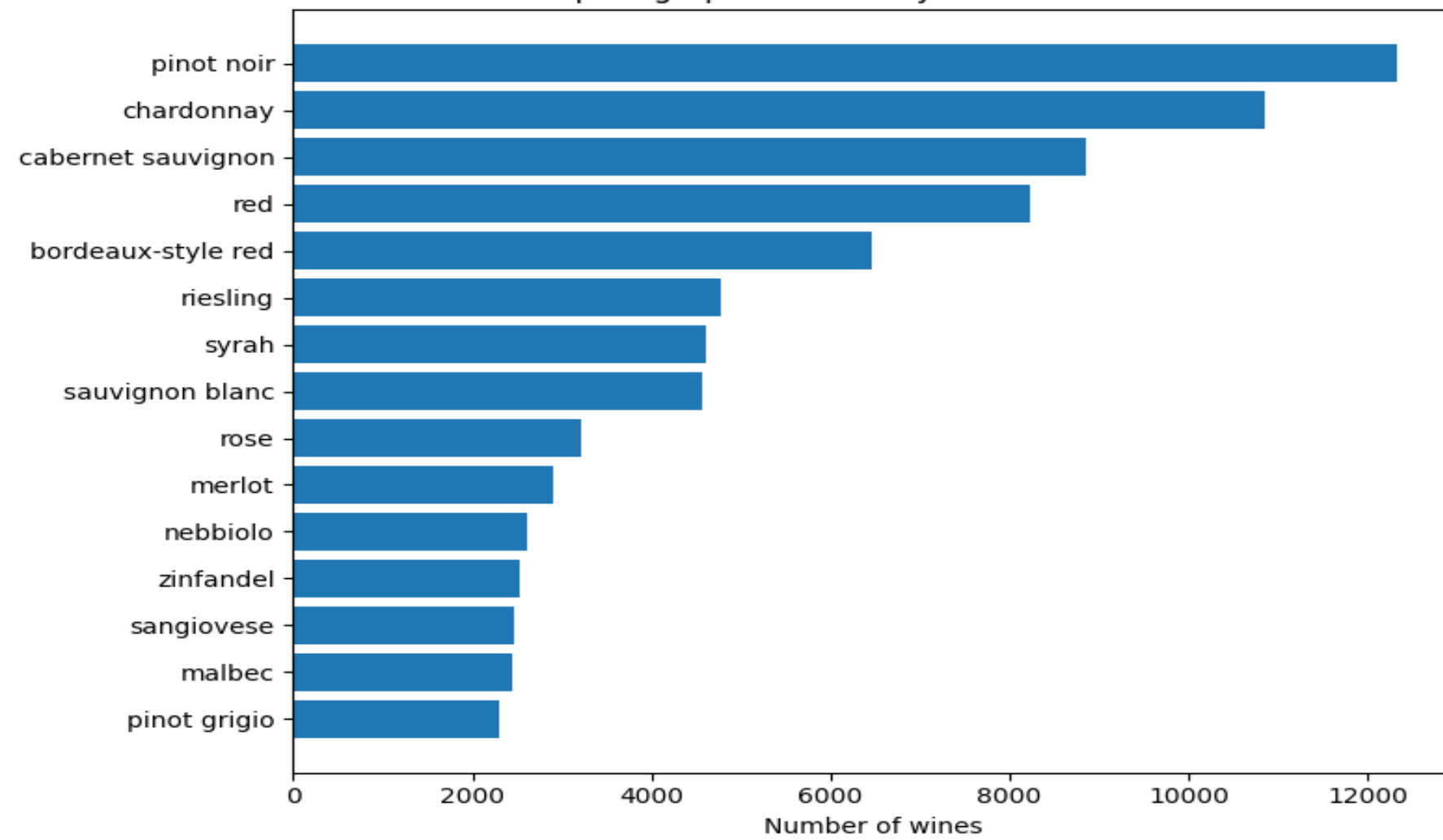


Top countries by average score (n_wines >= 500)

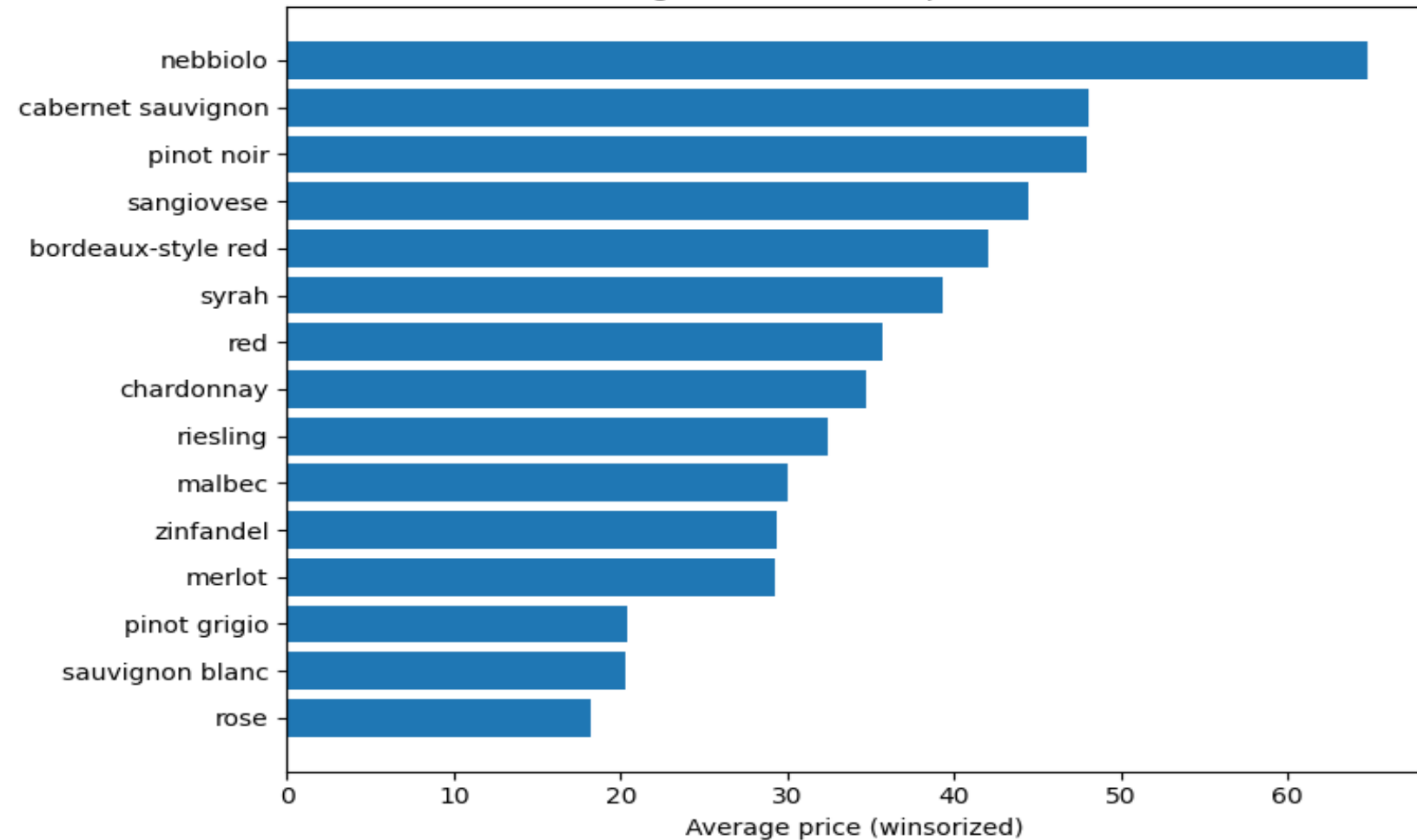


Overall, price varies strongly across countries, while score varies only by roughly 1 to 2 points across major producers.

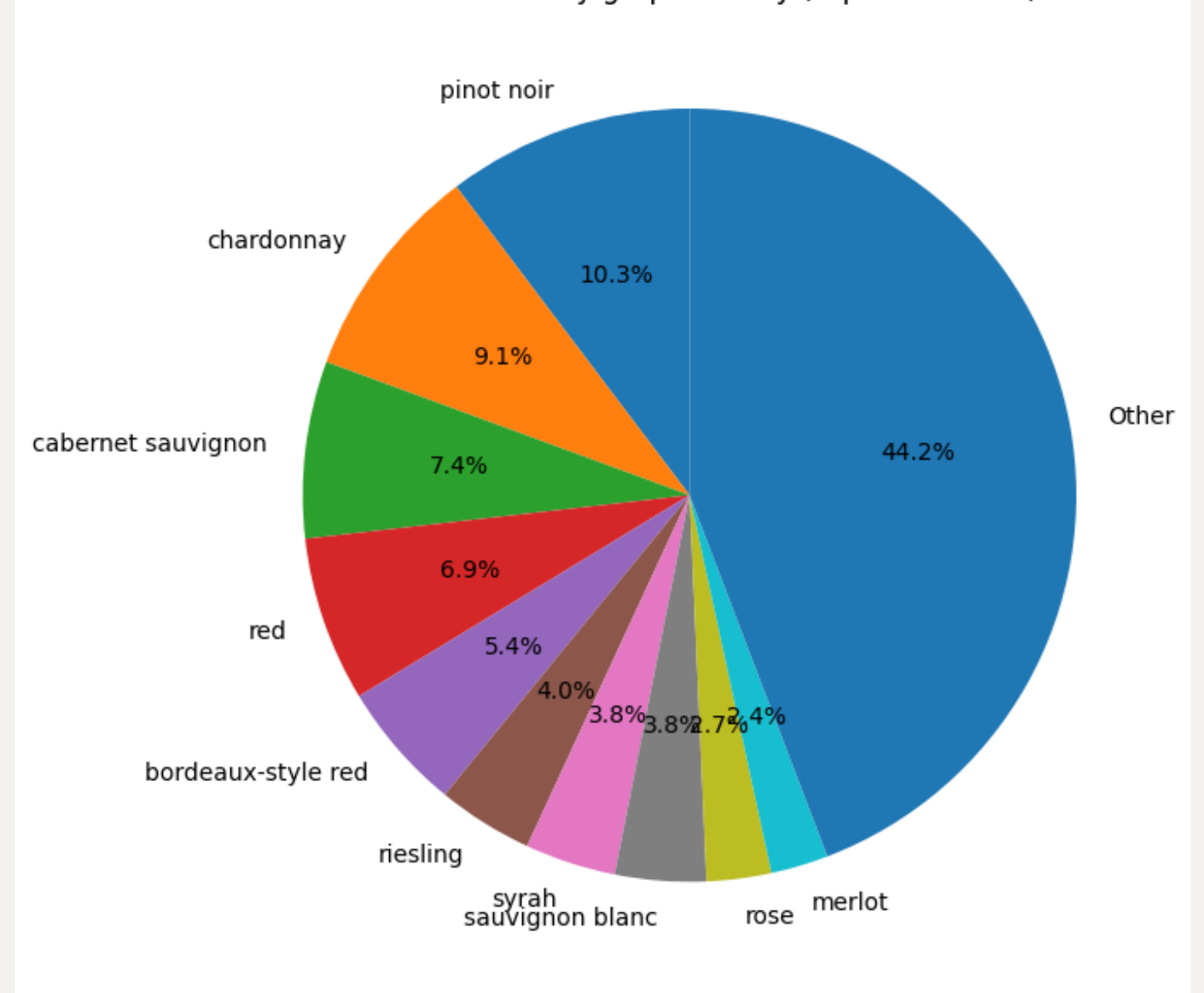
Top 15 grape varieties by number of wines



Top 15 grape varieties by average price
(among the 15 most frequent varieties)

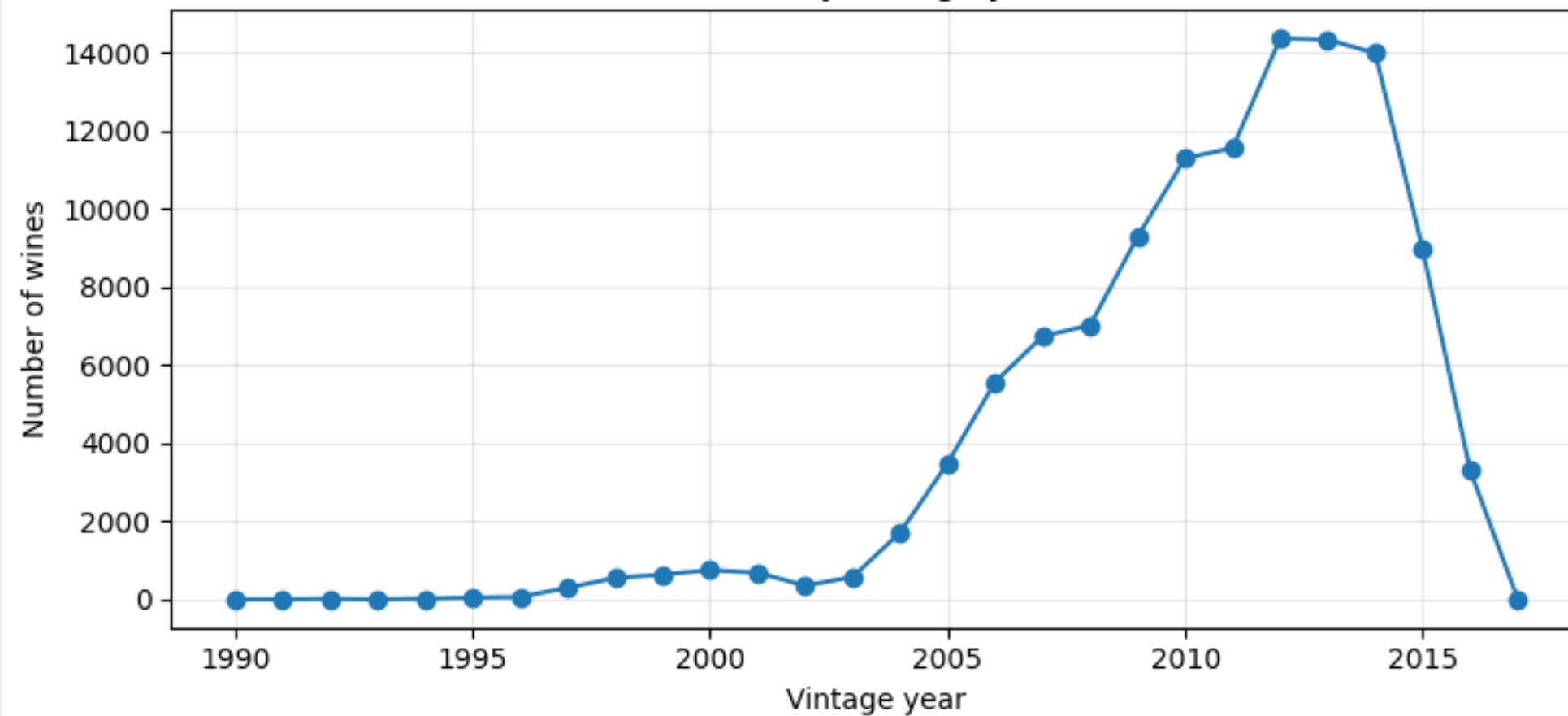


Share of wines by grape variety (Top 10 + Other)

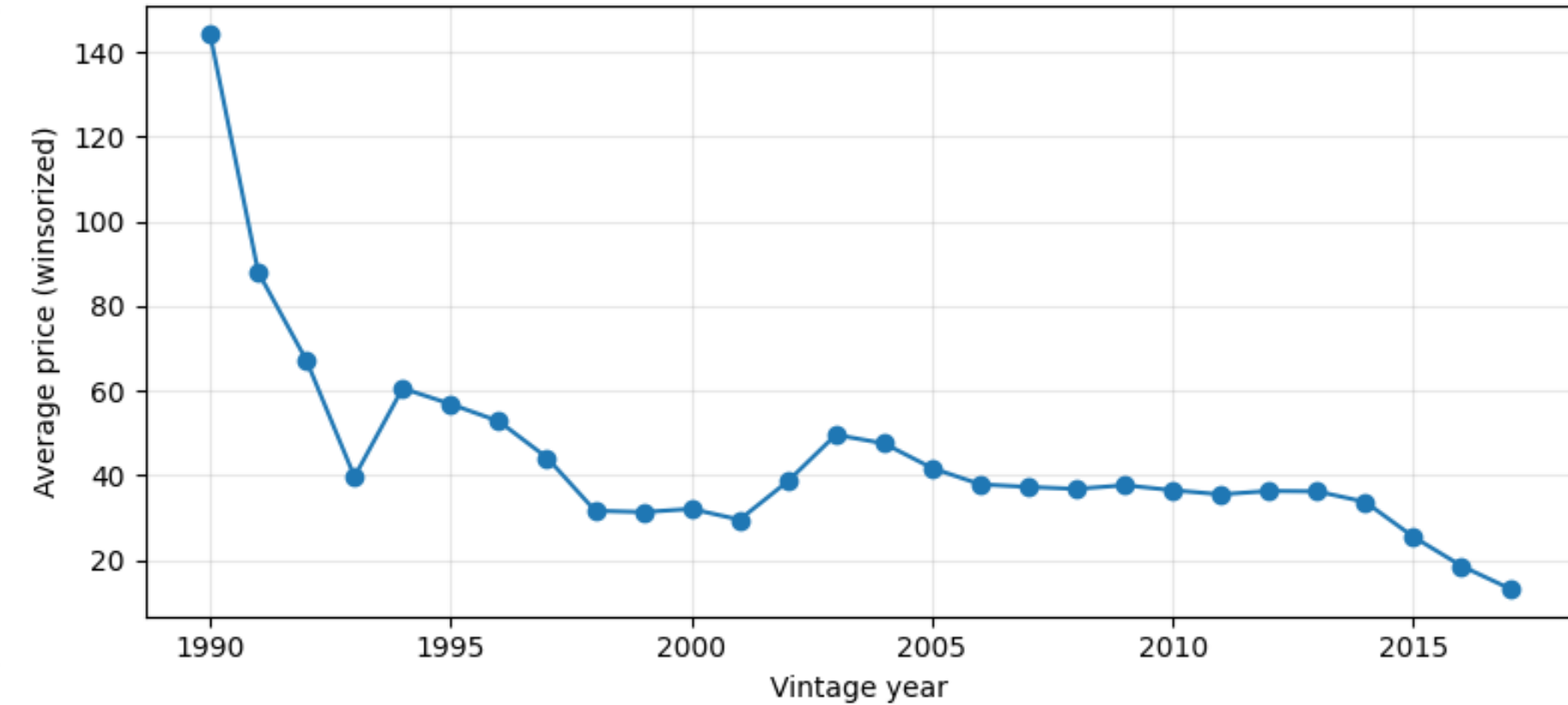


**Grape variety carries strong market and quality signals
that meaningfully influence downstream modeling.**

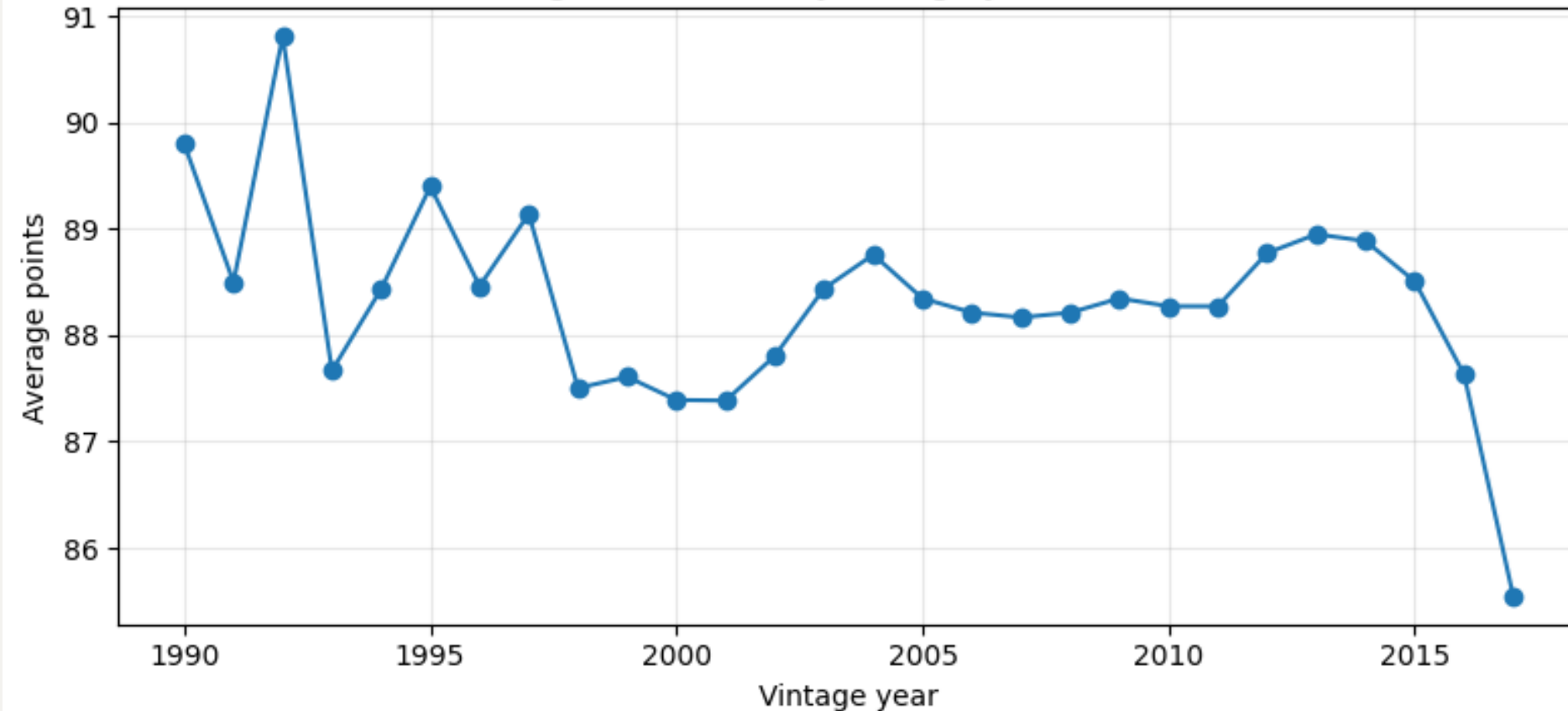
Number of wines by vintage year (≥ 1990)



Average price by vintage year (≥ 1990)

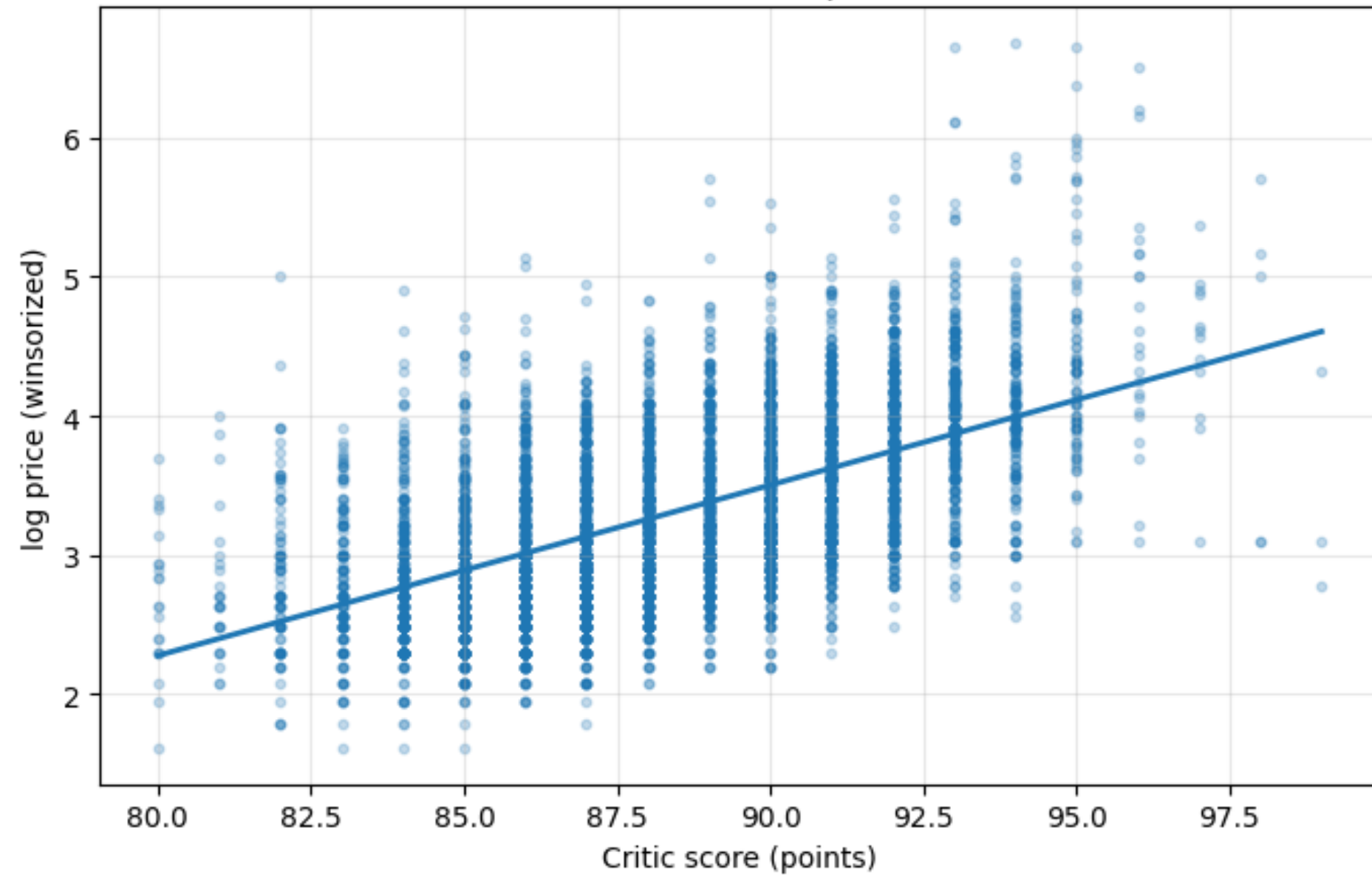


Average critic score by vintage year (≥ 1990)

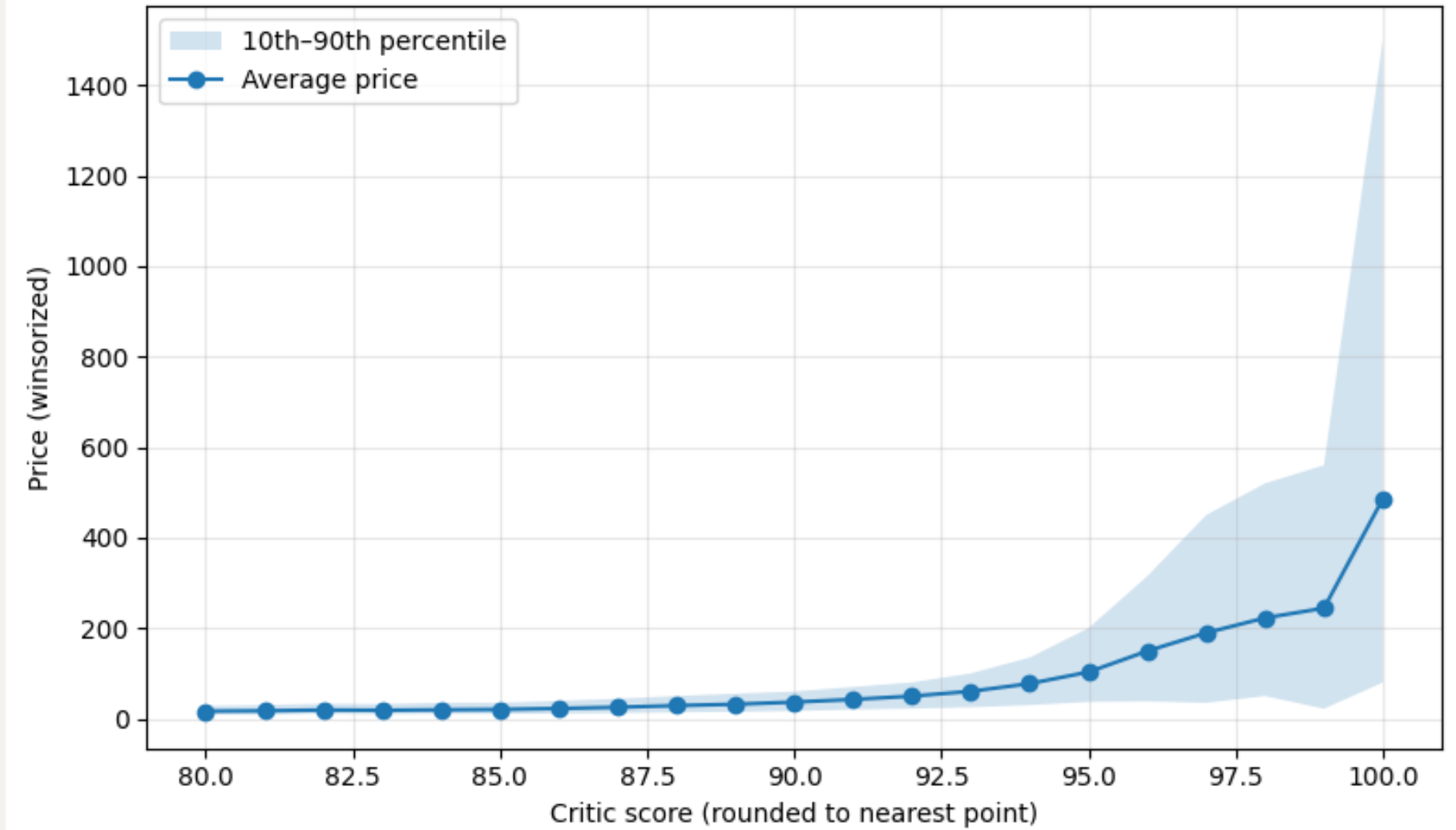


The vintage distribution reveals that the dataset is strongly skewed toward recent years.

Log-price vs. critic score
(corr = 0.594, fitted slope ≈ 0.1225)



Price distribution by critic score bin



The association between critic score and price is clearly positive but far from deterministic.

Recording:

https://psu.mediaspace.kaltura.com/media/DS+410+Final+Project+Recording/1_jy862alu